

# MRO Data Transport, Management & Accountability Challenges & Solutions

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**[Abstract]** The Mars Reconnaissance Orbiter (MRO) project is delivering unprecedented rates and volumes of science data from Mars; many times more than all previous Mars missions combined. This fact along with planned and unplanned constraints imposed by the flight and ground systems, has provided unique mission challenges in the areas of end-to-end data transport, management and accountability. MRO is breaking new ground in the area of data accountability. At the science observation level, a novel “FedEx®”-like system has been designed and implemented to track individual telemetry products from ground-based predictions, through on-board execution, downlink, product assembly, and availability to science centers. A web-based interface with hierarchical information display makes this tool useful for both low-level anomaly investigations, and user-level science product progress monitoring. A dedicated end-to-end Data Accountability Team (eeDAT) was formed to monitor and account for science data from science observation planning through delivery of resulting telemetry products to the science facilities. The team began handling a high volume of science data at the beginning of MRO’s Primary Science Phase in November of 2007, and currently handles approximately 10,000 individual science products per month. Along the way, the team has uncovered and addressed many issues in the end-to-end data system, made continuous improvements to team processes and tools, and accumulated lessons learned for future missions.

## I. Introduction

THE Mars Reconnaissance Orbiter (MRO) project is delivering unprecedented rates and volumes of science data from Mars; many times more than all previous Mars missions combined. This fact along with planned and unplanned constraints imposed by the flight and ground systems, has provided unique mission challenges in the areas of end-to-end data transport, management and accountability.

MRO utilizes a telemetry protocol stack based on Consultative Committee for Space Data Systems (CCSDS) standards, including the CCSDS File Delivery Protocol (CFDP) for handling of complete science products from spacecraft instrument to scientist, CCSDS Packet Telemetry to break up and transport segments of the science products, and CCSDS Advanced Orbiting Systems (AOS) Transfer Frames to reliably deliver data across the space link. MRO also employs a frame-level retransmission strategy designed to efficiently recover data that is lost in the space link. This paper describes how MRO uses the CCSDS protocol stack to its advantage, as well as how MRO has dealt with some of its limitations. Details of the retransmission strategy, rationale, and experiences are also covered.

The MRO telemetry stream is composed of interleaved science data from multiple instruments and engineering data of various types. Science data is inherently high rate and high volume, with end-to-end latency requirements on the order of 24 hours. Engineering data, on the other hand, is lower rate and lower volume, but must be delivered in a timely manner to support critical spacecraft operations. Aspects of both the flight and ground system architectures play a role in meeting the requirements for both classes of data. This paper details some of the challenges encountered and solutions implemented along the way to a solution that meets all requirements.

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MRO is breaking new ground in the area of data accountability. At the science observation level, a novel “Fed-Ex®”-like system has been designed and implemented to track individual products from ground-based predictions, through on-board execution, downlink, assembly, and availability to the science centers. A web-based interface with hierarchical information display makes this tool useful for low-level anomaly investigations, as well as user-level science product progress monitoring. MRO is also endeavoring to provide full accounting for all science data losses, whether at the File, Packet or Frame level. A dedicated end-to-end Data Accountability Team (eeDAT) was formed to address these challenges.

This paper describes MRO’s most significant data transport, management, and accountability challenges, and how MRO continues to address them. The paper also describes many lessons that have been learned along the way, the changes we have made as a result, and things to consider for future missions.

## II. MRO Science Lifecycle & Science Data Flow

Acquisition of MRO science can be described in terms of a life-cycle as depicted in Fig. 1. This diagram and accompanying description provides a simplified, conceptual view of the life-cycle for the purposes of this paper. Figure 2 is a simplified representation of the end-to-end Science Telemetry Data Flow.

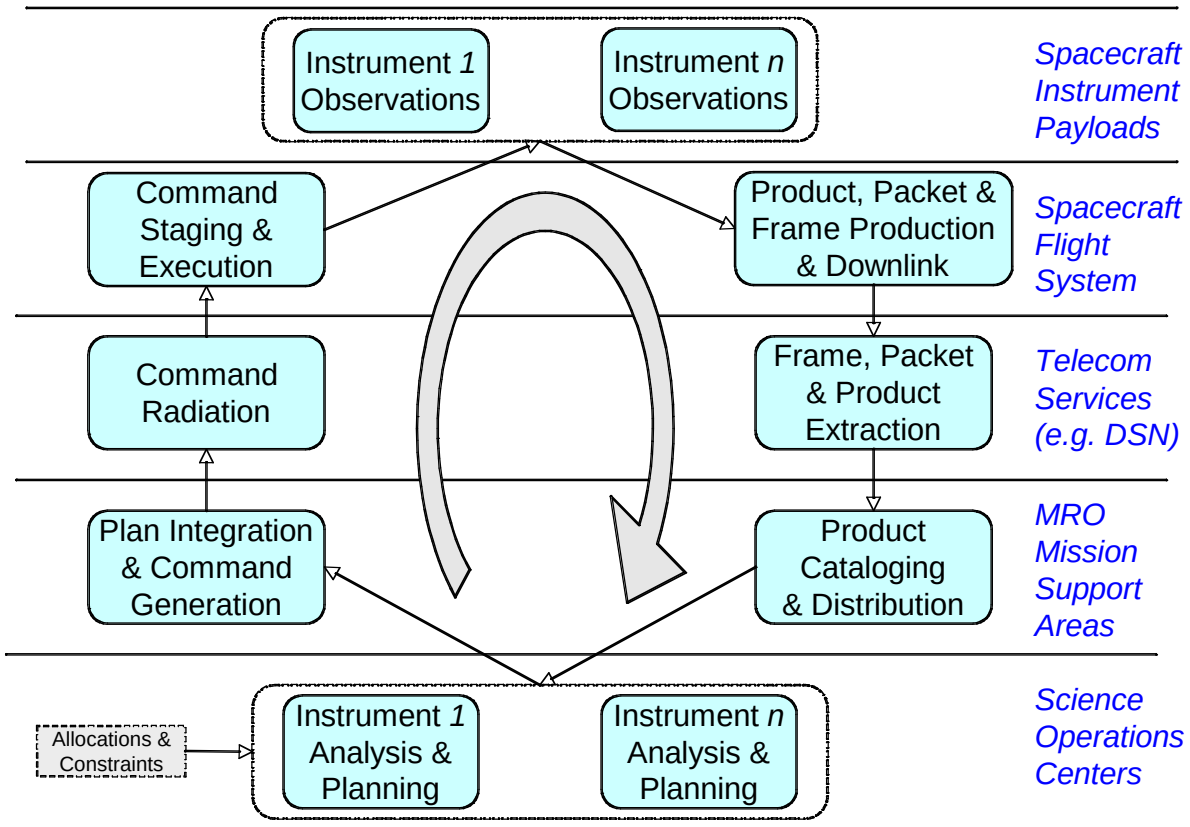
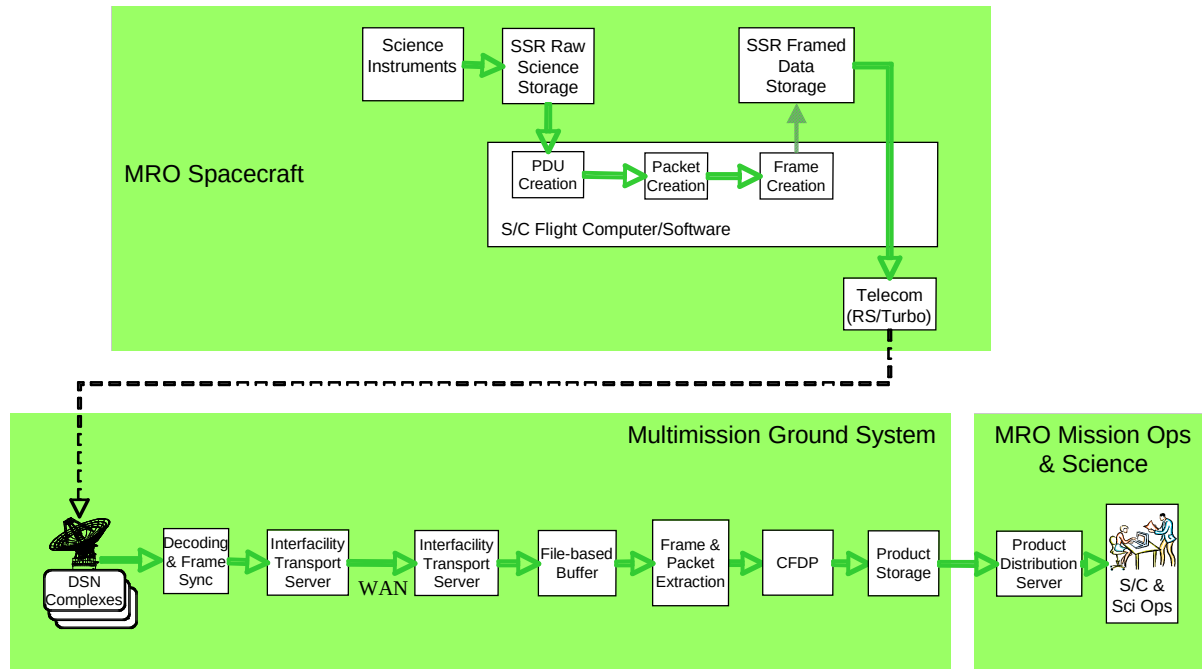


Figure 1. MRO science life-cycle



**Figure 2. Science telemetry data flow**

The following paragraphs describe each of the life-cycle steps in Fig. 1, with Fig. 2 providing the associated data flow context.

**Allocation & Constraints:** Each cycle begins when instrument teams receive allocation and constraints for the next planning cycle. One of the more significant allocations is the amount of downlink data volume allocated for each instrument.

**Instrument Analysis & Planning:** Given the results from prior planning cycles, and allocations and constraints for the current planning cycle, each instrument team comes up with a tentative list of desired observations along with priorities.

**Plan Integration & Command Generation:** Various MRO teams, including the instrument team and the spacecraft engineering team, then follow a formal process of integrating inputs, resolving conflicts, and generating spacecraft command files. One of these command files, the Integrated Target List (ITL), provides a list of observations and execution times.

**Command Radiation:** Command files are radiated to the spacecraft using the Deep Space Network (DSN).

**Command Staging & Execution:** Command files are staged onboard the spacecraft in advance of the time they are required. The spacecraft executes observations according to the schedule in the ITL file. For some instruments, the spacecraft issues the detailed commands necessary to accomplish the desired observation. For other instruments, the spacecraft passes an instrument-specific sequence file to the instrument that controls the detailed execution steps for the desired observation.

**Instrument Observations:** Instruments conduct the desired science observations and generate one or more telemetry data sets that are stored in the spacecraft Solid State Recorder (SSR).

**Product, Packet & Frame Production & Downlink:** Spacecraft flight software receives telemetry data sets from the instruments via the SSR. After a data compression step used by some instruments, the flight software encapsulates the data sets according the downlink protocol stack. Each telemetry data set is treated as a CFDP Product. Products are further packaged into CFDP Protocol Data Units (PDUs), then CCSDS Packets, then CCSDS AOS Transfer Frames. Observation data sets of the various instruments are produced asynchronously, and protocol encapsulation of multiple products may be conducted in parallel, resulting in production of a multi-instrument stream of Frames that are interleaved in time order. These frames are stored back to the SSR and then downlinked in first-in-first-out order.

**Frame, Packet & Product Extraction:** Frames are received by the DSN at the antenna complexes near Goldstone (USA), Canberra (Australia), and Madrid (Spain). Equipment at these complexes extracts Transfer Frames from the downlink stream, and routes them back to JPL via wide area networks. Data from each complex is accumulated into files, each nominally containing 30 minutes worth of telemetry received at the complex. Files from all complexes

are then processed in the order they are produced to extract Frames and Packets. Science packets, which contain PDUs, are then processed to produce CFDP Product files and metadata. Note that these Product files are identical to the telemetry data sets produced by the instruments, after compression where applicable. As files are produced, they are automatically transferred to our product cataloging and distribution system.

*Product Cataloging & Distribution:* Product and metadata files are received by the Raw Science Data Server (RSDS) which is responsible for cataloging products, formatting them for distribution, and serving them to distributed science operations centers. The science centers retrieve data via a multi-platform client, produce higher-level science products, and feed back the results to their upcoming planning cycles.

## **A. Product-based Science**

The MRO project chose to provide a CFDP Product-based service for delivery of science data to users versus delivery of a packet stream. The key reasons are: (a) Most science instruments conduct “observations” that produce finite, discrete data sets that naturally match with a product-based scheme; (b) Product Identifiers (IDs) can be associated with observations during the science planning stage, and these IDs can be tracked throughout the science data lifecycle at a high-level, without needing to understand the mapping between observations, packets, and frames; (c) A custom product-based protocol was successfully used by the Mars Exploration Rover (MER) project.

MRO uses the CFDP “unacknowledged” mode for telemetry. Unacknowledged mode was chosen based on analysis showing that the latency involved in closing the retransmission loop would often exceed the time from original downlink to the time data was no longer available onboard due to being overwritten by newer data. The most significant factor in the latency analysis was ground processing bandwidth constraints that could lead to up to 24 hours of ground latency alone. At MRO’s high rates, on-board overwrite times can be as short as three hours. Instead of product level retransmission, MRO uses frame-level retransmission. Since frame-level accountability can be done at the DSN tracking station, much of the ground latency associated with science data delivery and product building can be avoided. The retransmission process is described in detail later in this paper.

MRO found it necessary to customize CFDP in a few areas. Without customization, CFDP nominally releases products upon receipt of the “End PDU” for a product. This would not work for MRO because the DSN does not guarantee delivery of PDUs to the CFDP processing software in the same order as they were transmitted from the spacecraft. The reason is that data transfer from a DSN complex to JPL can become backlogged (for many hours), and then a new DSN complex can begin tracking and delivering data to JPL in near-real-time, jumping ahead of the first complex’s data. A second issue stems from MRO science users wanting/needing an early look at their products, even if they are not yet completed, to support timely decision-making. There are several scenarios where early product delivery is needed, including (a) The product is very large, could not be fully downlinked before a gap in DSN coverage, and the first part of the product is needed for decision making; (b) The entire product was downlinked, but one or more PDUs were lost, and the product is needed before the nominal CFDP product inactivity timer expires. Note that MRO would need to set this inactivity timeout to a very large value in order to wait for PDUs recovered by the frame-level retransmission process. To address these issues, two modifications were made to the MRO implementation of the CFDP specification: (1) MRO does not send an “End PDU”, which would normally trigger product delivery, but instead sends a “Metadata PDU” containing the product length. This allows the CFDP processing to use product completeness, instead of an End PDU, to deliver complete products; (2) A layer of software called File Delivery Manager (FDM) was developed to work in conjunction with vanilla CFDP to provide a variety of product release options. These options are summarized in Table 1.

| Timer                     | Description                                                                                                                                                                                                                                                                                                              | (Re)Starts when...                                                                           | Pays off when...                                        | PSP Value                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------|
| Inactivity Timer          | Designed to "give up" on a product after no new pieces have been received for a long time. Global=>Single setting applies to all instruments.                                                                                                                                                                            | Restarts whenever a PDU is received for a given product.                                     | No new PDU has been received within the timeout period. | 48 hours                                     |
| History Timer             | Determines what to do with PDUs received within a timeout period after a Transaction was closed. If the PDU is within the timeout period, it is thrown away, otherwise a new transaction with the same ID is opened. Global=>Single setting applies to all instruments.                                                  | A product is closed due to Product Completion, Cancel Directive, or Inactivity timeout.      | Timeout value has elapsed.                              | 48 hours                                     |
| Global Downlink Timer     | Designed to pause the inactivity/ history timers during lengthy interruptions in product data flow, such as between non-adjacent DSN passes. Global=>Single setting applies to all instruments.                                                                                                                          | Restarts whenever any PDU for any product is received.                                       | No new PDU has been received within the timeout period. | Disabled                                     |
| Quick-look Timer          | Optionally used to trigger early delivery of a partially received product. Can provide early indication of product reception, and/or access to key information at the beginning of a product. Does not close the transaction, so global triggers are still in effect. Instrument-specific.                               | First PDU is received for a given product.                                                   | Timer value has elapsed.                                | <varies by instrument; can also be disabled> |
| Latency Timer             | Optionally used to provide periodic delivery of partial products until processing for that product is done. Does not close the transaction. Will suppress delivery of duplicate products if nothing new is received, however Inactivity Timer will still deliver a final version even if unchanged. Instrument-specific. | Starts when Quick-look timer expires (if used), or when first PDU is received. Auto repeats. | Each timer timer value has elapsed.                     | <varies by instrument; recommend 4 hrs>      |
| Trigger                   | Description                                                                                                                                                                                                                                                                                                              |                                                                                              |                                                         |                                              |
| Complete Product Recived  | Transactions are nominally closed normally immediately after receipt of one copy of all constituent PDUs.                                                                                                                                                                                                                | n/a                                                                                          | n/a                                                     | n/a                                          |
| Cancel Directive          | eeDAT can ask DSOT to issue a Cancel Directive that closes processing of a product, and delivers whatever has accumulated as a partial product. Only the History Timer remains in effect.                                                                                                                                | n/a                                                                                          | n/a                                                     | n/a                                          |
| Abandon Directive         | eeDAT can ask DSOT to issue an Abandon Directive that closes processing of a product, and discards whatever has accumulated. No product is delivered. Only the History Timer remains in effect.                                                                                                                          | n/a                                                                                          | n/a                                                     | n/a                                          |
| Version Release Directive | eeDAT can ask DSOT to issue a Version Release Directive that will essentially force expiration of the Quick-look or Latency timer, whichever would occur first. Product delivery and Timer behavior is just as if time had advanced to that point.                                                                       | n/a                                                                                          | n/a                                                     | n/a                                          |

**Table 1. MRO product release extensions to CFDP**

### III. The eeDAT Team

In recognition of the challenges presented by the unprecedented MRO science rates and volumes, MRO management made a strategic decision to establish a dedicated end-to-end Data Accountability Team (eeDAT). Some key challenges faced by this new team are:

- (1) Very High Data Rates & Volumes (for a planetary mission). The maximum telemetry bit rate is 6 Megabits/sec, the maximum daily volume is 230 Gigabits/day, and the expected total mission volume is more than 34 Terabits. Note also that these volumes represent raw, often compressed level-0 science data. Besides keeping up with these rates under nominal circumstances, the team had to be able to keep up during off-nominal situations, including being able to catch up following unplanned interruptions in data flow.
- (2) First Use of CFDP on a Mars Mission. Although the Mars Exploration Rover mission used a custom file-based protocol, MRO was the first Mars mission to use an adaptation of the CCSDS standard CFDP file-based protocol. Several mission factors resulted in visibility, latency, and quality challenges associated with file-based telemetry. Among these were: file sizes up to 28 Gigabits, one-way light times up to 20 minutes, limited on-board storage capacity, and losses in the downlink.
- (3) Complex Flight/Ground Data Processing. Simultaneous handling of X-band and Ka-band telemetry streams, Turbo and Reed-Solomon encoding, packetizing and framing well in advance of downlink, multiple virtual channels with different contents and priorities, and frame-level retransmission.
- (4) Bandwidth Constraints and Prioritization Issues. MRO science rates and volumes exceed the ability of our end-to-end telemetry system to deliver in real-time, whereas our relatively low-rate engineering telemetry must be delivered in near-real-time. This results in prioritization challenges at various points in the telemetry system.

The eeDAT team was purposely established early in the mission development phase so that the team would have an opportunity to impact the design and implementation of both the flight and ground systems in order to facilitate monitoring at strategic points in the science data lifecycle.

The responsibilities of the eeDAT team can be summarized as follows:

- Serve as the single point of contact for the project's product data accountability
- Provide end-to-end science and relay data product accountability, from science observation/product planning (by science users), through science product delivery (to science users)
- Provide status of science data products at any point in their delivery paths
- Monitor performance of end-to-end data product flow and provide periodic reporting to users and service providers
- Diagnose and coordinate resolution of end-to-end data flow problems, including those related to quantity, quality, completeness, and latency
- Maintain cognizance of the planning process and generate predicted science data outcomes
- Provide frame-level retransmission scheduling, performance monitoring, and problem resolution
- Coordinate activities of Multimission Data Management and Science Data Services teams

The eeDAT team designed and implemented a combination of tools and procedures to help meet these responsibilities. One of the key tools is a Fed-Ex®-like system for tracking science data throughout the science data lifecycle. This tool is described in detail in a later section of this paper. Formal procedures were written to cover the following key team activities:

- (1) Product Prediction Procedure – describes how the team receives inputs from the science planning process, and generates and manages product predictions associated with those plans.
- (2) Missing Product Procedure – describes how the team recognizes the case where a product was expected but was never received, how to determine the cause, and how to follow-up.
- (3) Partial Product Procedure - describes how the team recognizes the case where an incomplete product has been received, how to determine the cause, and how to follow-up.
- (4) Periodic Summary Reporting Procedure – identifies periodic reports that the team generates, and describes when and how they are generated and distributed.
- (5) Retransmission Procedure – describes how the team manages a frame-level retransmission process.
- (6) System Monitoring & Maintenance Procedures – describes various procedures used to ensure that the systems and tools that support team activities remaining running properly.
- (7) Quick Reference Procedure – describes various low-level tricks-of-the trade that are routinely used by the team on an as-needed basis.

The eeDAT team currently consists of several part-time individuals, each of whom also has responsibilities on other MRO teams. Membership on other teams helps ensure good inter-team communications, and provides the eeDAT team with key insights when diagnosing mission system-wide issues. Team members were also chosen to represent a variety of backgrounds and areas of expertise allowing the team to conduct special in-depth analyses and studies. In addition to the various formal team procedures described above, team members provide direct user support on a weekly rotating basis. The on-duty member sits on-console during normal business hours, and can be reached via phone after hours. The team meets approximately every other day to share status and discuss issues.

eeDAT has successfully supported the mission throughout each of its operations phases, including Launch, Cruise, Mars Orbit Insertion, Aerobraking, and the first several months of the Primary Science Phase.

#### **IV. Science Data Tracking**

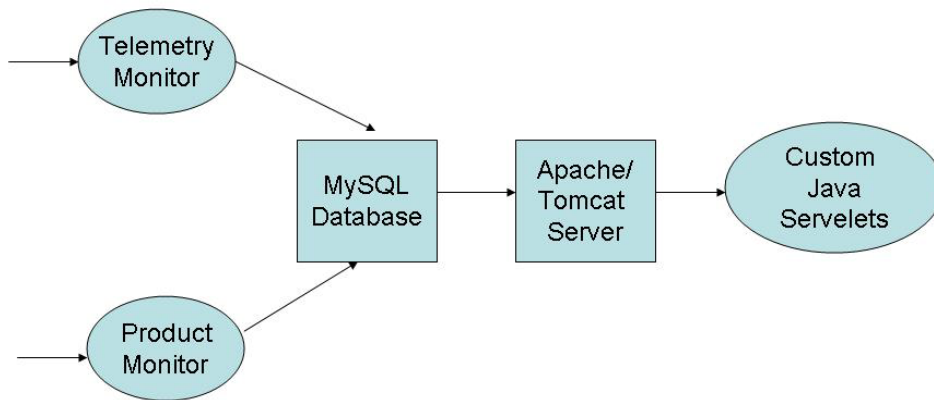
MRO is now in its primary science phase and produces between 2000 and 3000 science products each week. To track all these products MRO Project members currently rely on the Tailorable Report Utility for Science Product Tracking, aka TRUST. The idea behind TRUST was to provide users with a relatively simple web interface that shows where products currently are in the data pipeline, similar to the web interface that FedEx® offers to customers for tracking their packages in the FedEx® system. TRUST provides users with continuously updated information on

- a) What products were planned?
- b) What products were commanded?
- c) What products were created onboard the spacecraft?
- d) What products have been received on the ground?

e) Which products were complete, which were partial and which are currently missing?

### A. Tracking Science Products

Every week the MRO science planning team takes input from the different payload teams and creates an integrated list of products that will be produced on the spacecraft the following week. The information from this list of predicted products is loaded into the TRUST database. If someone makes a mistake, there are procedures and tools to remove products from the database. Meanwhile, two processes are running continuously and updating the TRUST. One process monitors what products are being created and placed on the project database. The other process looks at spacecraft telemetry in realtime. The TRUST developers worked with MRO flight software early on to ensure that engineering telemetry would be sent to indicate strategic milestones in the spacecrafts creation of a science product. Both these programs will add status information to the TRUST database to show where any of the predicted products are in the data pipeline. These monitoring programs feed data into a MySQL® database. Custom java servlets allow users to access the database info via an Apache/Tomcat Web Server.



**Figure 3. Processes supporting TRUST**

TRUST users specify a time range and get a time-ordered list of all MRO products predicted for that range, ordered by their predicted spacecraft event time. Each instrument onboard MRO is unique; the points tracked by TRUST do not always apply to all MRO instruments, and can mean different things for different instruments. TRUST offers users a variety of ways to filter this product view so that users can view just the products they are interested in.

Each row in Fig. 4 shows information on an individual product. The product ID is at the far left, followed by the instrument, the Spacecraft Event Time (SCET) when the science observation for that product was predicted to begin, and the Earth Received Time (ERT) when the product was finally complete on the ground. The orange, green and yellow columns give status information on the product as it is received by the TRUST monitors. The orange column indicates that the product is planned and command files have been sent. The green column means there are indicators in the telemetry that the command for creating the product has occurred onboard the spacecraft. The yellow column indicates that the product has been created onboard, is available in the Project Distributed Object Manager (DOM) Database and finally is available in the RSDS science repository. If TRUST has received any of these indicators for a given product, it marks the appropriate column with “====” to report the indicator has been received. For the yellow columns reporting on the project database and the science repository, a value of “==” would indicate that only part of the expected science product was received. If a given indicator is not received, the column remains blank.

From this list, users can click on an individual product. This takes them to another web page with more detailed information on the product selected.

[Product Filter](#)
[Logout](#)
[Help](#)

Update Page

Records: 4387

TRUST Product Selection & Status

Current Since: 2007/111-00:16:57

Apply Filter

Instrument:

ALL

Start From :

Now

1

3

10

30 Days ago

End At :

Now

1

3

10

30 Days from now

☐ Exclude Complete Products

| Products                        | Instruments | Predict Planned Product Start Date | Last ERT          | PRED  | SEQ   | RAD | ONB | ICMD  | DLNK | TLM   | DOM   | RSDS  | EEDAT |
|---------------------------------|-------------|------------------------------------|-------------------|-------|-------|-----|-----|-------|------|-------|-------|-------|-------|
| <a href="#">4A-020000517B00</a> | CRISM IR    | 2007/091-10:39:13                  | 2007/091-21:36:18 | ===== | ===== |     |     | ===== |      | ===== | ===== | ===== |       |
| <a href="#">4A-030000517B00</a> | CRISM VNIR  | 2007/091-10:39:13                  | 2007/091-21:36:26 | ===== | ===== |     |     | ===== |      | ===== | ===== | ===== |       |
| <a href="#">4A-05100E800600</a> | MARCI       | 2007/091-10:47:01                  | 2007/092-06:29:39 | ===== | ===== |     |     | ===== |      | ===== | ===== | ===== |       |
| <a href="#">4A-014831BCF700</a> | HIRISE      | 2007/091-11:02:30                  | 2007/092-08:22:47 | ===== | ===== |     |     | ===== |      | ===== | ===== | ===== |       |

Figure 4. Listing of science products in TRUST

Figure 5 shows more detailed information on a single product – in this case 4A-014830153200, a High Resolution Imaging Science Experiment (HiRISE) product. These fields display the various indicators the TRUST monitors have received on this particular product. Additionally, the white window in the center contains information on spacecraft event records (EVRs) that were created onboard the spacecraft within a time range surrounding the products predicted spacecraft event time. EVRs could indicate other problems onboard the spacecraft that might effect the creation of this particular product.

Product Filter   Product Select & Status   Custom Query   Logout   Help

Update

MRO SPACECRAFT ONBOARD STATUS: 4A-014830153200

Current Since: 2007/113-02:11:06

|                               |                       |                                 |                       |                                |                       |
|-------------------------------|-----------------------|---------------------------------|-----------------------|--------------------------------|-----------------------|
| <b>Onboard Command Status</b> |                       | <b>Onboard Execution Status</b> |                       | <b>Onboard Downlink Status</b> |                       |
| Verification:                 | 2007/081-15:08:48.691 | Inst SCET:                      | 2007/083-04:59:24.166 | Pred Dlink UTC:                | 2007/083-22:14:26     |
| Validation Status:            | Success               | Data/SSR UTC:                   | 2007/083-07:05:14.249 | CFDP Dlink Start Time:         | null                  |
| CRC Status:                   | 0xf400d324            | Data/SSR Size:                  | 652,499,964 bytes     | CFDP Dlink End Time:           | 2007/083-05:26:16.197 |
| Filename:                     | 1211110706.mod        | Pred Exe UTC:                   | 2007/083-04:58:11     |                                |                       |
|                               |                       | Pred Prod End:                  | 2007/083-04:58:27     |                                |                       |
|                               |                       | Pred Prod Size:                 | 632,708,500 bytes     |                                |                       |

FORWARD LINK

|                         |                       |                                     |       |                        |                                       |
|-------------------------|-----------------------|-------------------------------------|-------|------------------------|---------------------------------------|
| <b>Sequence/Command</b> |                       | <b>Info, Errors &amp; Anomalies</b> |       | <b>CFDP Processing</b> |                                       |
| Last:                   |                       | SCET                                | ID    | Description            | File:                                 |
| Onbrd File:             | d:/hir/1211110706.mod | 083-08:36:22.415                    | 19997 | IN_TD_MISSED_TGT_WINDO | 4A-014830153200-2007-083T09.42.31.dtl |
|                         |                       |                                     |       |                        | Last UTC:                             |
|                         |                       |                                     |       |                        | 2007/083-09:42:31                     |
|                         |                       |                                     |       |                        | Last Size:                            |
|                         |                       |                                     |       |                        | 652,499,964 bytes                     |
|                         |                       |                                     |       |                        | Cmplt:                                |
|                         |                       |                                     |       |                        | 100 %                                 |
|                         |                       |                                     |       |                        | File State:                           |
|                         |                       |                                     |       |                        | COMPLETE                              |
|                         |                       |                                     |       |                        | Fill:                                 |
|                         |                       |                                     |       |                        | 0                                     |
|                         |                       |                                     |       |                        | Trans State:                          |
|                         |                       |                                     |       |                        | FINISHED                              |
|                         |                       |                                     |       |                        | Timer State:                          |
|                         |                       |                                     |       |                        | Closed Normally                       |
|                         |                       |                                     |       |                        | Gap Count:                            |
|                         |                       |                                     |       |                        | 0                                     |

PRODUCT PREDICTION

|               |                                |
|---------------|--------------------------------|
| Latest UTC:   | 2007/081-16:51:36              |
| Coortl ID:    | MH-SCI                         |
| Compression:  | ENABLE                         |
| Obs Type:     | Small Off-Nadir High Stability |
| Latitude:     | -27.222                        |
| Longitude:    | 325.453                        |
| Orbit Number: | 3077a                          |
| Request Pri:  | 5                              |
| RF Band:      | X                              |

RSDS PROCESSING

|                  |                   |
|------------------|-------------------|
| Latest from DOM: | 2007/083-09:49:55 |
| Latest to Users: | 2007/083-10:20:35 |

RETURN LINK

Figure 5. Individual product information



## B. When data is not received ...

A large majority of the thousands of products that MRO produces every week are received without incident. However, MRO has taken the unprecedented step of tracking and validating *all* losses of science data. If you go back to the first screen in TRUST (Fig. 4), you'll notice that the column on the far right is labeled "eeDAT". When a product is missing, an eeDAT member investigates why, and adds that information to the TRUST database, where all users can access the explanation. eeDAT initially attempted to do this with all partial products as well. However, investigating the number of partial products proved impractical; instead eeDAT now monitors and investigates frame gaps in telemetry received. Since a single incident of rain in Spain can cause many partial products, and since retransmission can mask the occurrence of frame gaps, it was felt that daily frame accountability gave the project a better view of how the space-link and ground system were performing. (The various causes of frame loss are described later in this paper.) A product that was completely missing was judged serious enough to require a direct investigation.

Not all problems are going to show up in the TRUST view as partial or missing products. Occasionally, the instrument payloads on the spacecraft produce products that are not on the predicted products list. eeDAT affectionately terms these "BONUS" products. While bonus products do not show up in the standard TRUST display, their appearance in the processing pipeline is noted by the TRUST monitors and information on them is recorded and stored in the TRUST database, where it can be accessed by eeDAT as a special case.

The table below represents product accounting during the time period between Jan 1<sup>st</sup>, 2007 and July 9<sup>th</sup> 2007.

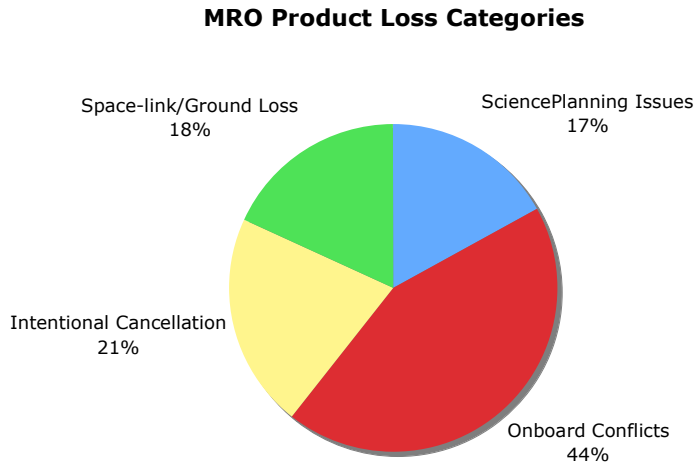
| <b>Instrument</b>                                                                      | <b>Complete</b> | <b>Partial</b> | <b>Missing</b> | <b>Bonus</b> |
|----------------------------------------------------------------------------------------|-----------------|----------------|----------------|--------------|
| High Resolution Imaging Science Experiment (HIRISE)                                    | 1189            | 490            | 94             | 2            |
| Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) – Infrared (IR)           | 9783            | 688            | 709            | 55           |
| Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) – Visible & Near Infrared | 10128           | 314            | 733            | 57           |
| Context Imager (CTX)                                                                   | 2830            | 319            | 138            | 39           |
| Mars Color Imager (MARCI)                                                              | 1970            | 318            | 84             | 28           |
| Shallow Radar (SHARAD)                                                                 | 2751            | 450            | 189            | 1            |
| <b>TOTAL</b>                                                                           | <b>28651</b>    | <b>2579</b>    | <b>1947</b>    | <b>182</b>   |

(Note: The bookkeeping for Mars Climate Sounder (MCS) products is a little different than the other instruments and eeDAT is still developing reporting tools that will handle MCS products. MCS products account for almost ½ of the total science products, but less than ¼ of 1% of the total science data volume for MRO. MCS data is routinely downlinked 3 times so the overall loss of MCS data has been negligible.)

MRO science products are missing for a variety of reasons. Some things can go wrong in the early planning stages. Science teams can use incorrect product ids, or the software that builds the integrated product list can have bugs. Problems have occurred onboard that have prevented some products from being created. Of course problems have occurred in the space-link that have prevented products from being received at the Deep Space Network. eeDAT has investigated things like performance during lower elevation angles to improve the number of products returned. The Deep Space Network also has its share of hardware, software and planning problems that have caused some loss of science products. There have been problems in moving data between the DSN and JPL that have caused the loss or delay of products. Once the products have been created at JPL, there have been some problems in moving them out to where they can be accessed by the science teams. During this time period there were also many cases where individual science Teams decided to nullify their observation commands at the last minute so the product shows up in this table as missing, even though it was not expected. They do this by replacing the real instrument commands with a no-operation (NO-OP) command. The causes of missing products can be categorized as:

| <b>Explanation</b>                                   | <b># Missing Products</b> |
|------------------------------------------------------|---------------------------|
| Problems in science planning process                 | 329                       |
| Conflicts onboard the spacecraft                     | 851                       |
| Products intentionally cancelled by Science Teams    | 418                       |
| Problems related to the space-link and ground system | 349                       |

The relative magnitudes of lost products associated with each of these categories are illustrated in Fig. 6.



**Figure 6. Product loss categories**

The entry of the spacecraft into safe-mode in March of 2007 accounts for the majority of the cases of onboard conflicts. The causes of partial products have been exclusively space-link and ground system related problems.

### **C. The Missing Product Investigation**

During the Operational Readiness Test prior to the beginning of the primary science phase, it became clear that the requirements for data accountability were not completely understood. We had a good start with TRUST and the DSN's new frame-accountability capability, but if we were going to be able to account for every science product planned and/or produced for MRO, we needed additional tools for a) monitoring products and b) analyzing data to determine why a product was not produced.

On the surface, the problem looked simple. However, that first question "Have we lost frames?" in an anomalous situation was much more involved than was originally thought. The DSN had a new capability to detect and report frame gaps at the station and some bugs were to be expected. The eeDAT team also developed an additional gap detection tool that is run against the frames that are successfully transferred to JPL and stored in the frame-level database. This was needed to 1) double-check the reports from the station for accuracy, and 2) identify losses in the transfer of data to JPL, or JPL processing.

But if we are considering products, we need to know if frames were lost while a particular product was being downlinked. The Project knows when the products should have been created – all products had a predicted Spacecraft Event Time (SCET). But there was no clear answer to what the Earth Receive Time (ERT) for a product should be. Because of the flow of realtime, recorded and retransmitted data, the correlation between the ERT and SCET is constantly changing on the ground. What we needed was a capability to correlate events on the spacecraft with events (frame loss) on the ground. We accomplished this with two different tools. The first tool looked at an ERT ordered list of transfer frames and extracted the first SCET value if it actually existed in the frame. The second tool took a list of products as they were expected to be created onboard (SCET ordered), and printed that out along with the ERT of any of the complete or partial products. Since we know something about the order in which products will be downlinked, this information could be used to estimate a time period when a given missing product should have been downlinked.

If no frame loss was detected during the estimated downlink time for a missing product, then we had to consider that a problem might have occurred on the spacecraft or in the science planning process. The spacecraft team already has a daily report on what happened onboard – including all instrument commands executed. This was very useful, but was also fairly lengthy and took some effort to digest. However, there were a known set of Event Records (EVRs) that flight software would produce that indicated a problem with the creation of a science product. eeDAT set up another daily report to look for these records on any given day.

If frame loss was detected during the estimated downlink time for a missing product, an eeDAT member would then actually sit down and examine the reports from the DSN to correlate what DSN events were responsible for large frame gaps. Periods where there were numerous small gaps were characterized as “ratty” and were also noted and explained if possible.

These different daily reports were summarized in a single report that would go out to all MRO Project members, covering

- a) A gap report summary
- b) A list of significant events at any of the DSN stations monitoring MRO
- c) A summary of science products received in the science repository
- d) A listing of product-related spacecraft event records, taken from engineering telemetry

This report provided a good daily snapshot of what was happening in the end-to-end data flow, but tracking data accountability for all MRO products remains a fairly labor-intensive process. The process has been worthwhile in that eeDAT has uncovered several significant bugs in the processing of telemetry data at JPL and is currently investigating some frame loss at the station where the cause is unknown.

## **V. Quantity & Performance Metrics**

This section provides and describes various quality and performance metrics from MRO’s primary science phase. These metrics are typical of those that are regularly produced and reported by the eeDAT team. The following metrics are covered:

Frame-level totals – Downlinked, captured, missing, and retransmitted.

Causes of frame loss.

Product-level totals – Planned, produced, captured (partial & complete), missing, and bonus.

Causes of missing, partial & bonus products.

### **A. Frame-level Totals Metrics**

The eeDAT team uses automated daily scripts to query all frames from our frame-level database, and produce raw daily statistics including number of original frames received, number of original frames missing, number of retransmitted frames received, and number of gaps. On a weekly basis, eeDAT imports this information into a spreadsheet and produces reports by day, week, and month. Figs. 7 and 8 are an example of a weekly summary table and a plot of total telemetry volume relative to important mission milestones, respectively.

| Planning Week                   | DOY Range    | Original Downlink (VC6) |                       |                       |           |           | Retransmission (VC6)    |                             |                        |                         | Summary (VC6)       |  |
|---------------------------------|--------------|-------------------------|-----------------------|-----------------------|-----------|-----------|-------------------------|-----------------------------|------------------------|-------------------------|---------------------|--|
|                                 |              | Expected (# of frames)  | Present (# of frames) | Missing (# of frames) | # of Gaps | % in Gaps | Recovered (# of frames) | Still Missing (# of frames) | % of Missing Recovered | % of Expected Recovered | Total (# of frames) |  |
| 001 1a                          | 2006-312-315 | 19,590,154              | 18,509,329            | 1,080,825             | 16,551    | 5.52      | 234,519                 | 846,306                     | 21.70                  | 1.20                    | 18,743,848          |  |
| 001 1b                          | 2006-316-322 | 42,143,308              | 41,718,534            | 424,774               | 14,880    | 1.01      | 321,093                 | 103,681                     | 75.59                  | 0.76                    | 42,039,627          |  |
| 002 1a                          | 2006-323-329 | 45,916,676              | 45,372,833            | 543,843               | 2,692     | 1.18      | 15,275                  | 528,568                     | 2.81                   | 0.03                    | 45,388,108          |  |
| 002 1b                          | 2006-330-336 | 38,154,732              | 37,925,231            | 229,501               | 5,691     | 0.60      | 201,702                 | 27,799                      | 87.89                  | 0.53                    | 38,126,933          |  |
| 003 1a                          | 2006-337-343 | 37,568,545              | 37,490,874            | 77,671                | 4,498     | 0.21      | 50,011                  | 27,660                      | 64.39                  | 0.13                    | 37,540,885          |  |
| 003 1b                          | 2006-344-350 | 42,552,057              | 40,097,642            | 2,454,415             | 714       | 5.77      | 280,623                 | 2,173,792                   | 11.43                  | 0.66                    | 40,378,265          |  |
| 004 1a                          | 2006-351-357 | 57,017,344              | 50,441,553            | 6,575,791             | 20,978    | 11.53     | 33,199                  | 6,542,592                   | 0.50                   | 0.06                    | 50,474,752          |  |
| 004 1b                          | 2006-358-364 | 52,412,696              | 51,809,344            | 603,352               | 9,553     | 1.15      | 257,502                 | 345,850                     | 42.68                  | 0.49                    | 52,066,846          |  |
| 004 1c                          | 2006-365-006 | 53,864,294              | 53,436,137            | 428,157               | 7,013     | 0.79      | 263,741                 | 164,416                     | 61.60                  | 0.49                    | 53,699,878          |  |
| 005 1a                          | 2007-007-013 | 54,532,331              | 54,353,276            | 179,055               | 2,561     | 0.33      | 120,350                 | 58,705                      | 67.21                  | 0.22                    | 54,473,626          |  |
| 005 1b                          | 2007-014-020 | 33,416,624              | 32,723,042            | 693,582               | 11,466    | 2.08      | 529,674                 | 163,908                     | 76.37                  | 1.59                    | 33,252,716          |  |
| 006 1a                          | 2007-021-027 | 40,710,156              | 38,142,649            | 2,567,507             | 8,471     | 6.31      | 45,004                  | 2,522,503                   | 1.75                   | 0.11                    | 38,187,653          |  |
| 006 1b                          | 2007-028-034 | 49,706,585              | 49,435,283            | 271,302               | 10,892    | 0.55      | 47,852                  | 223,450                     | 17.64                  | 0.10                    | 49,483,135          |  |
| 007 1a                          | 2007-035-041 | 43,030,028              | 42,418,154            | 611,874               | 17,736    | 1.42      | 550,009                 | 61,865                      | 89.89                  | 1.28                    | 42,968,163          |  |
| 007 1b                          | 2007-042-048 | 50,542,509              | 47,318,108            | 3,224,401             | 433,430   | 6.38      | 1,086,115               | 2,138,286                   | 33.68                  | 2.15                    | 48,404,223          |  |
| 008 1a                          | 2007-049-055 | 59,933,364              | 59,127,209            | 806,155               | 18,093    | 1.35      | 363,202                 | 442,953                     | 45.05                  | 0.61                    | 59,490,411          |  |
| 008 1b                          | 2007-056-062 | 46,323,628              | 45,813,817            | 509,811               | 9,959     | 1.10      | 77,007                  | 432,804                     | 15.11                  | 0.17                    | 45,890,824          |  |
| 009 1a                          | 2007-063-069 | 56,055,888              | 54,913,198            | 1,142,690             | 2,066     | 2.04      | 743,128                 | 399,562                     | 65.03                  | 1.33                    | 55,656,326          |  |
| 009 1b                          | 2007-070-076 | 33,759,600              | 33,637,176            | 122,424               | 3,069     | 0.36      | 11,144                  | 111,280                     | 9.10                   | 0.03                    | 33,648,320          |  |
| 010 1a                          | 2007-077-083 | 23,081,876              | 23,049,224            | 32,652                | 408       | 0.14      | 10,096                  | 22,556                      | 30.92                  | 0.04                    | 23,059,320          |  |
| 010 1b                          | 2007-084-090 | 50,661,865              | 48,071,671            | 2,590,194             | 2,063     | 5.11      | 529,610                 | 2,060,584                   | 20.45                  | 1.05                    | 48,601,281          |  |
| 011 1a                          | 2007-091-097 | 68,520,559              | 64,313,497            | 4,207,062             | 26,474    | 6.14      | 1,680,499               | 2,526,563                   | 39.94                  | 2.45                    | 65,993,996          |  |
| 011 1b                          | 2007-098-104 | 61,397,616              | 59,357,331            | 2,040,285             | 2,357     | 3.32      | 858,262                 | 1,182,023                   | 42.07                  | 1.40                    | 60,215,593          |  |
| 012 1a                          | 2007-105-111 | 73,614,001              | 73,055,917            | 558,084               | 32,395    | 0.76      | 373,824                 | 184,260                     | 66.98                  | 0.51                    | 73,429,741          |  |
| 012 1b                          | 2007-112-118 | 67,614,881              | 67,216,502            | 398,379               | 88,030    | 0.59      | 700                     | 397,679                     | 0.18                   | 0.00                    | 67,217,202          |  |
| 013 1a                          | 2007-119-125 | 60,213,794              | 59,583,644            | 630,150               | 11,882    | 1.05      | 312,134                 | 318,016                     | 49.53                  | 0.52                    | 59,895,778          |  |
| 013 1b                          | 2007-126-132 | 61,324,469              | 61,109,783            | 214,686               | 29,618    | 0.35      | 150,521                 | 64,165                      | 70.11                  | 0.25                    | 61,260,304          |  |
| 014 1a                          | 2007-133-139 | 73,567,240              | 70,393,373            | 3,173,867             | 173,907   | 4.31      | 6,522                   | 3,167,345                   | 0.21                   | 0.01                    | 70,399,895          |  |
| 014 1b                          | 2007-140-146 | 70,389,651              | 69,972,502            | 417,149               | 37,436    | 0.59      | 3,524                   | 413,625                     | 0.84                   | 0.01                    | 69,976,026          |  |
| 015 1a                          | 2007-147-153 | 81,855,197              | 81,521,543            | 333,654               | 21,755    | 0.41      | 98,090                  | 235,564                     | 29.40                  | 0.12                    | 81,619,633          |  |
| 015 1b                          | 2007-154-160 | 94,143,661              | 77,639,756            | 16,503,905            | 252,636   | 17.53     | 1,010,133               | 15,493,772                  | 6.12                   | 1.07                    | 78,649,889          |  |
| 016 1a                          | 2007-161-167 | 72,655,540              | 72,014,372            | 641,168               | 95,443    | 0.88      | 74,165                  | 567,003                     | 11.57                  | 0.10                    | 72,088,537          |  |
| 016 1b                          | 2007-168-174 | 67,957,780              | 66,614,030            | 1,343,750             | 108,112   | 1.98      | 186,067                 | 1,157,683                   | 13.85                  | 0.27                    | 66,800,097          |  |
| 017 1a                          | 2007-175-181 | 68,272,065              | 63,147,493            | 5,124,572             | 151,588   | 7.51      | 7,019                   | 5,117,553                   | 0.14                   | 0.01                    | 63,154,512          |  |
| 017 1b                          | 2007-182-188 | 71,477,809              | 71,295,505            | 182,304               | 11,502    | 0.26      | 58,482                  | 123,822                     | 32.08                  | 0.08                    | 71,353,987          |  |
| <b>PSP Phase Totals To Date</b> |              |                         |                       |                       |           |           |                         |                             |                        |                         |                     |  |
| Frames (VC6)                    |              | 1,923,978,523           | 1,863,039,532         | 60,938,991            | 1,645,919 | 3.17      | 10,590,798              | 50,348,193                  | 19.80                  | 2.46                    | 1,873,630,330       |  |
| Gbits (VC6)                     |              | 16,952                  | 16,415                | 537                   |           |           |                         | 93                          | 444                    |                         | 16,509              |  |
| Frames (MCS)                    |              | 4,816,102               | 4,816,102             |                       |           |           |                         |                             |                        |                         | 4,816,102           |  |
| Gbits (MCS)                     |              | 42                      | 42                    |                       |           |           |                         |                             |                        |                         | 42                  |  |
| Frames (total)                  |              | 1,928,794,625           | 1,867,855,634         |                       |           |           |                         |                             |                        |                         | 1,878,446,432       |  |
| Gbits (total)                   |              | 16,995                  | 16,458                |                       |           |           |                         |                             |                        |                         | 16,551              |  |

Figure 7. Frame-level totals by week and to-date

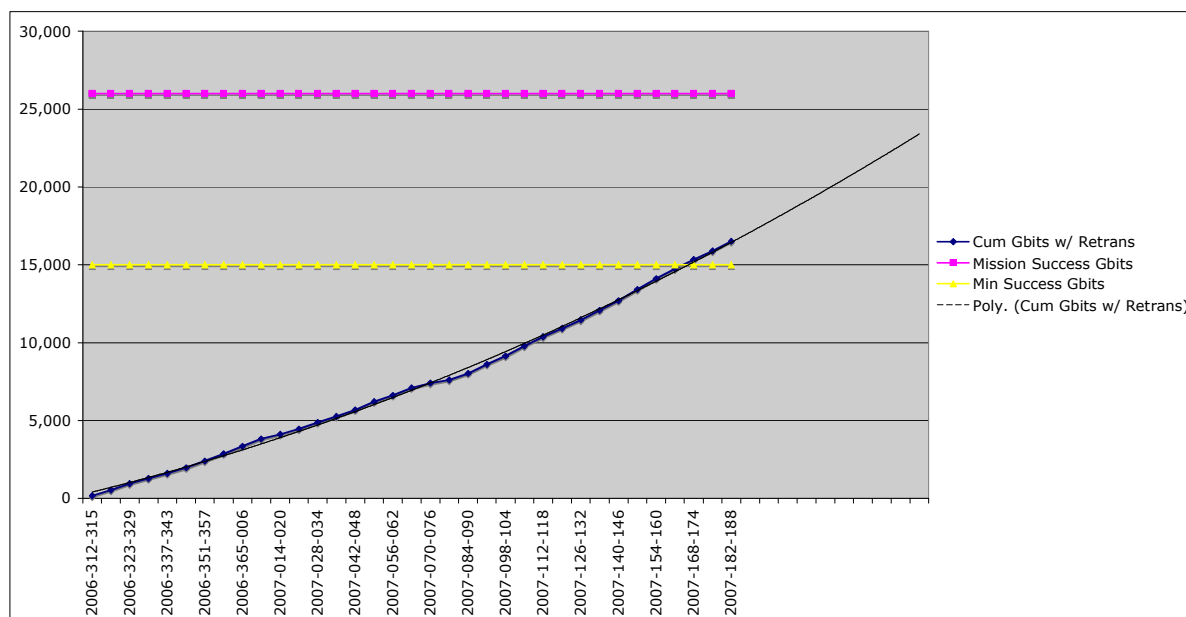
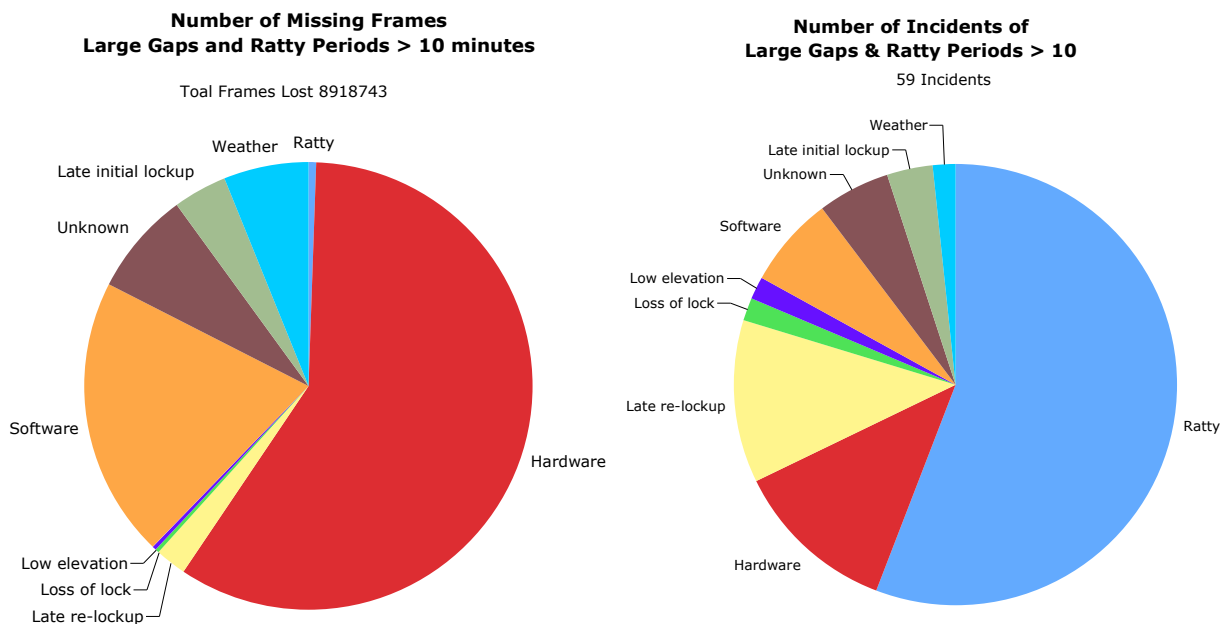


Figure 8. Total mission volume versus key milestone levels

## B. Causes of Frame Loss

The eeDAT automated daily scripts create both summary level and detailed information concerning gaps in the stream of telemetry frames. Gap detection is primarily based on the Virtual Channel Frame Counter (VCFC) field within the header of each frame. eeDAT follows a manual procedure each work day which attempts to identify the cause for any large gaps (greater than 10,000 frames), as well as any “ratty” periods characterized by many small gaps over a sustained period of time.

Figure 9 shows the major causes of frame-level gaps, both in terms of total number of frames missing, and total number of incidents.



**Figure 9. Major causes of frame-level losses (example for one month)**

The categories and their meanings are:

Hardware – includes DSN hardware/mechanical failures, power outages, spacecraft anomalies, ...

Software – frames were lost due to software anomalies

Late initial lockup – a new DSN station was unable to lock up on the spacecraft signal in the allotted time

Late re-lockup – a DSN station was not able to re-acquire the signal after a Mars occultation in the allotted time

Loss of lock – a DSN station was unable to maintain lock

Weather – bad weather causes a degraded signal-to-noise ratio resulting in intermittent data loss

Low elevation – atmospheric effects cause a degraded signal-to-noise ratio resulting in intermittent data loss

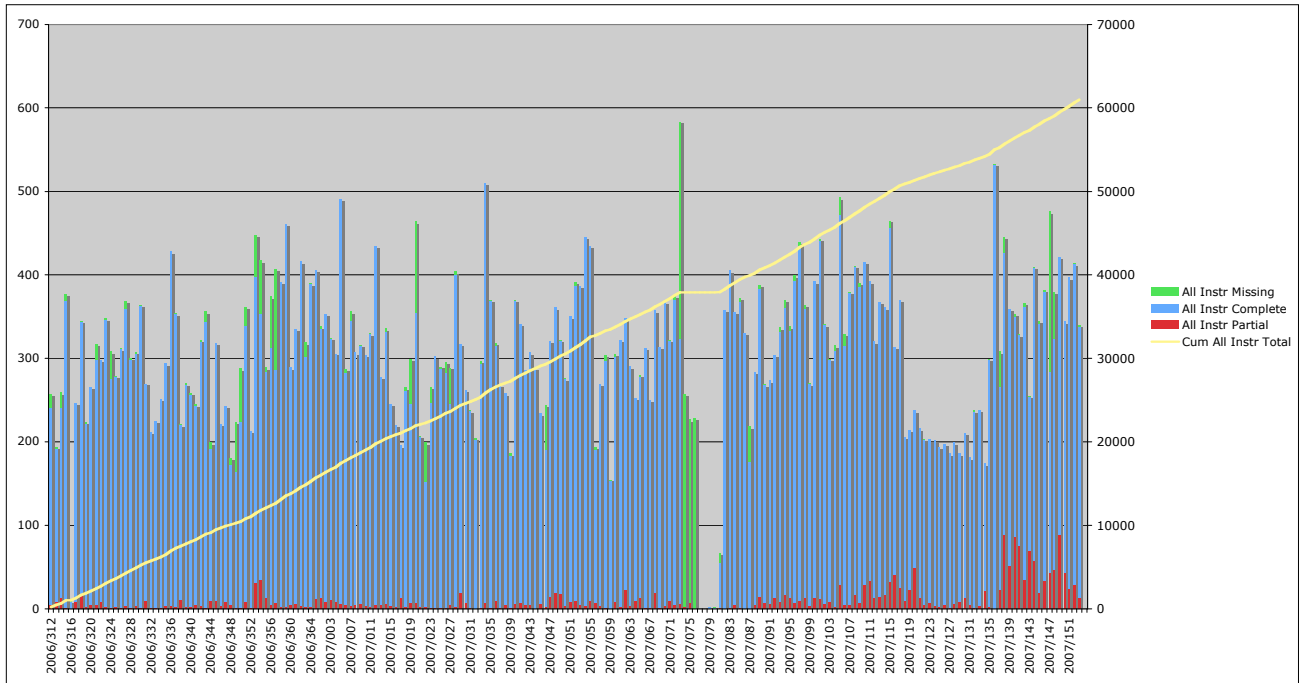
Ratty – a period of intermittent data loss that is not clearly attributable to low Signal-to-Noise Ratio (SNR)

Unknown – the cause of the frame loss could not be determined or is still under investigation

Note that when considering the total number of lost frames, the “hardware” category dominates. This result has been consistent since the start of the primary science phase. By contrast, when considering the number of incidents of frame loss, we see that the “Ratty” category dominates, even though it was a very small contributor to the total number of lost frames. Although these charts represent only a single month of operations, these results are typical of all months. Also note that a period of rattiness lasting 10 minutes or more is considered an event.

## C. Product Level Totals Metrics

Figure 10 is an example of product-level metrics. This example provides stacked bars for each day of the primary science phase, for partial, complete, and missing products, as well as a line chart showing the cumulative number of products received to date.



**Figure 10. Product-level totals metrics**

This diagram is useful for seeing trends in product production and in product quality. For example, the partial product count (red) is trending higher toward the end of the graph. A special investigation was initiated to understand and resolve this issue. This investigation is ongoing, but the losses appear to be coincident with moving to higher telemetry rates. The higher rates forced us to use a less powerful error correction code, and decreased the effectiveness of frame-level retransmission since the on-board data overwrite times were decreased.

#### **D. Causes of Missing and Partial Products**

As was illustrated in Fig. 6, the dominant cause of missing products was losses in the space-link or on the ground. A closer examination at the frame level has revealed that these losses are primarily due to DSN tracking issues, either caused by significant hardware failures or difficulties with locking up on the MRO signal.

The dominant cause of partial products has been “ratty” periods of telemetry. These ratty periods can be caused by adverse weather conditions at the tracking stations, and/or low elevation atmospheric effects. We also have a predictable level of rattyness in our Turbo data flows due to a known spacecraft anomaly that cannot be corrected in flight. Finally, as we have gone to higher telemetry rates and stand-alone Reed-Solomon encoding, we have been observing regular ratty periods that were not previously observed. eeDAT is currently leading an effort to identify the cause, and potential solutions for this new behavior.

## **VI. Retransmission**

The Mars Reconnaissance Orbiter spacecraft uses an enhanced subset of the CCSDS CFDP protocols to build and downlink data products, but does not utilize the corresponding uplink portion of the protocol. In order to fill in lost frames not received in the initial downlink, the MRO project utilizes a system which identifies missing frames not received by each DSN tracking station and auto-generates a command file to be uplinked by an MRO ACE which requests specific frames or ranges of frames to be retransmitted by the MRO spacecraft from its onboard Solid State Recorder (SSR.) Numerous enhancements and changes have been made to the system of software programs and procedures that support the MRO retransmission process: this portion of the paper will discuss some of the highlights of this process.

#### **A. Rationale for MRO data retransmission**

MRO uses X-band frequencies relayed through the DSN series of antennas, similarly to prior US Mars spacecraft. However, MRO has an unprecedented downlink data-rate of up to 6 Megabits/sec. Previous US probes to Mars have sent back data at a maximum data rate of only 80 kilobits/sec (Mars Global Surveyor) and 110 kilobits/sec (Mars Odyssey), meaning MRO is sending back data at over 50 times higher rates. To take advantage of this high bandwidth, many of the observations taken by MRO are quite substantial in size, which means that many of MRO's products are sizable targets for the occasional loss of telemetry inherent in space communication. For example, to receive a complete full-sized HiRISE image of 28 Gigabits streaming down at over 90 data packets per second for well over an hour (when at 6 Megabits/sec) requires the space link to be *perfect* for well over an hour. Lower data rates, of course, may challenge the telemetry system a bit less but provide more time for factors like wayward rain clouds, the occasional DSN equipment problem, earth-based or space-based radio-frequency interference (RFI), or other factors to cause data dropouts.

A delicate balance between having acceptable link margins and high data rates is constantly in play throughout the MRO observation planning process, especially when considering that link margin varies due to earth weather, space weather, elevation angle of the receiving DSN antenna, and even varies due to small performance variations between similarly-sized DSN antennas. Since a central feature of MRO mission success is the successful delivery of 26 Terabits or more of science data, there is a powerful motivation to try and maximize data return at every opportunity.

The underlying data-encoding schemes, which provide substantial error-correction, eliminate many of the small downlink data corruptions, but they have their limits. At higher data rates the most-powerful error-correction encoding schemes cannot be used due to terrestrial spectrum usage agreements for X-band communication which limit us to a maximum of 6 Mega-symbols/sec, and DSN performance limitations. This leads to a squeeze on data quality from two directions: the higher data rates have a higher chance of having small data quality problems, but the most-powerful data error-correction schemes are not available at those same higher data rates!

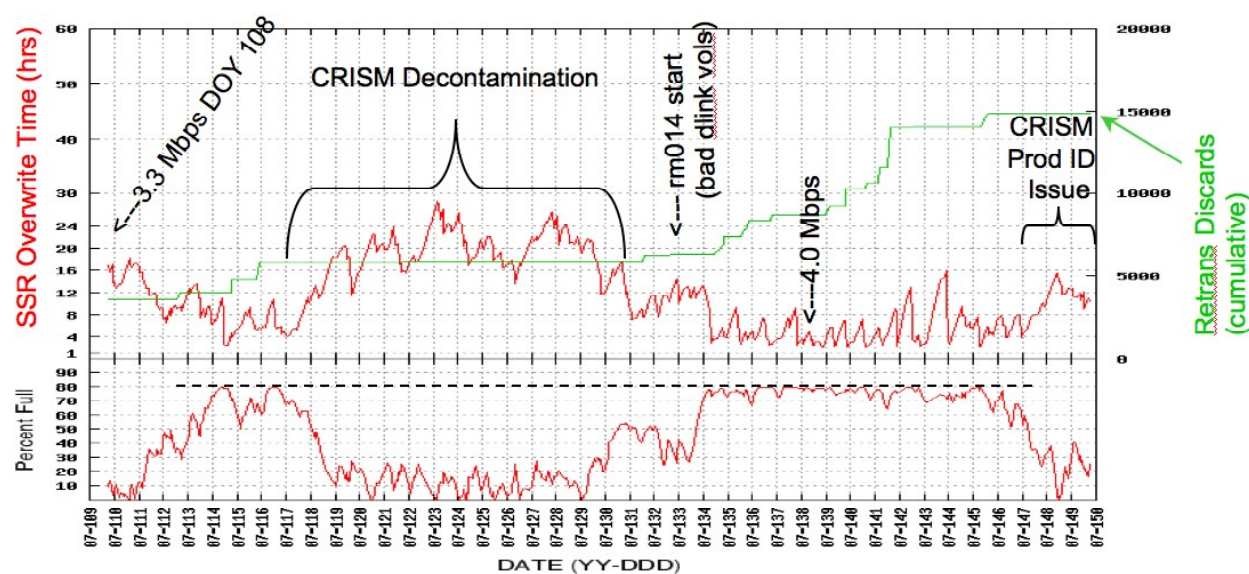
## **B. Methods and procedures of MRO retransmission**

A suite of tools to facilitate retransmission was developed for MRO, both inside and outside of the project. The facilitating outside development is a system that generates Frame Accountability Reports (FARs) at each DSN station as data is received. These FARs indicate the frame number range of data received over five-minute periods. If a gap is noticed, indicating frames missing in transmission, the FARs note this. The frame numbers are placed on the data by the MRO spacecraft, and the frame counter rolls over in the 16-million-frame range. Clever programming allows MRO to identify each missed frame.

The internally-developed tools “rs” and “rccron” allow MRO personnel and scripts to build a weekly crontab file, used to call the “rcgen” program which builds retransmission commands. A brief overview:

1. **“rs”** (standing for “retransmission scheduler”) takes spacecraft event files, light-time files, ACE on-console schedule files, and a variety of user-specified switches to build a retransmission schedule file which includes: start and stop times for FARs of interest, the date/time to run the cron job, the type of telemetry encoding, and other parameters. This schedule file is used as input to “rccron”.
2. **“rccron”** takes as input the schedule file, configuration files, and other parameters to build a correctly-formatted crontab file which is installed on a UNIX workstation. This crontab file calls “rcgen.”
3. **“rcgen”** is the workhorse program which, when run, generates retransmission request files and passes them to an automated command generation process, which in turn produces binary command files that are ready for ACE radiation. “rcgen” also writes critical information to a log file and to online data storage areas to allow traceability and troubleshooting. In addition, a small shell script is run immediately after every crontabbed “rcgen” instance; this shell script collects important information including: run results, number of instruction lines, and number of frames in the just-submitted command input; it mails this info to several e-mail accounts. This script informs the ACE on-console what the outcome of the “rcgen” run was. There are occasions when no gaps are found and therefore no command is built; without the info script, the ACE might be left sitting waiting for a command to load and radiate which will never be built. The script also notifies the science community if a “large” (greater than 200,000 frames) request has been submitted to the command-build system, so that they can adjust their science data-volume planning if necessary. Since retransmitted data has a higher priority than regular science data, a large retransmission may cause a backup of data onboard the spacecraft and the science teams need as much notice as possible to re-plan their observations. The e-mail messages are also sent to certain eeDAT cell phones using text messaging, which allows the eeDAT team members to stay informed about the success and size of retransmission command builds.

There are several timing factors that impinge on the retransmission process. The most-important mechanical one is the time-until-overwrite of data onboard MRO. The overwrite time varies due to many factors, including downlink rate, number and size of scheduled observations, and fullness of the onboard SSR in frame space (this last being where the products themselves are built onboard.) The overwrite time can vary from in excess of 24 hours to less than three hours. The availability of DSN coverage (more or fewer hours) influences how quickly products built onboard can be drained from frame space. Figure 11 illustrates the relationships between SSR fullness (bottom red-line values; 80% full corresponds to using 100% of available SSR frame space); time until on-board data overwrite (upper red-line value, units in hours); and cumulative retransmission instruction lines discarded due to their already having been over-written on-board.



**Figure 11. SSR fullness, overwrite time, and retransmission effectiveness**

Note that low percentage-full corresponds to more generous time-until-overwrite and also to very few retransmission instructions discarded, whereas high percentage-full corresponds to short overwrite time and larger number of discarded retransmission requests (less-successful retransmission performance.)

During the science planning process, the science teams must perform a delicate balancing act with respect to SSR storage. Their key considerations are: (a) they want to generate enough data volume so they don't waste downlink bandwidth (under-producing); (b) they want to avoid overflowing their on-board space allocation causing loss of observations (over-producing); and (c) they want to maintain a average fill-level that is low enough to result in a reasonably long time-before-overwrite so that retransmission can be most effective. Several software predictive tools, beyond the scope of this paper, are used for science data planning.

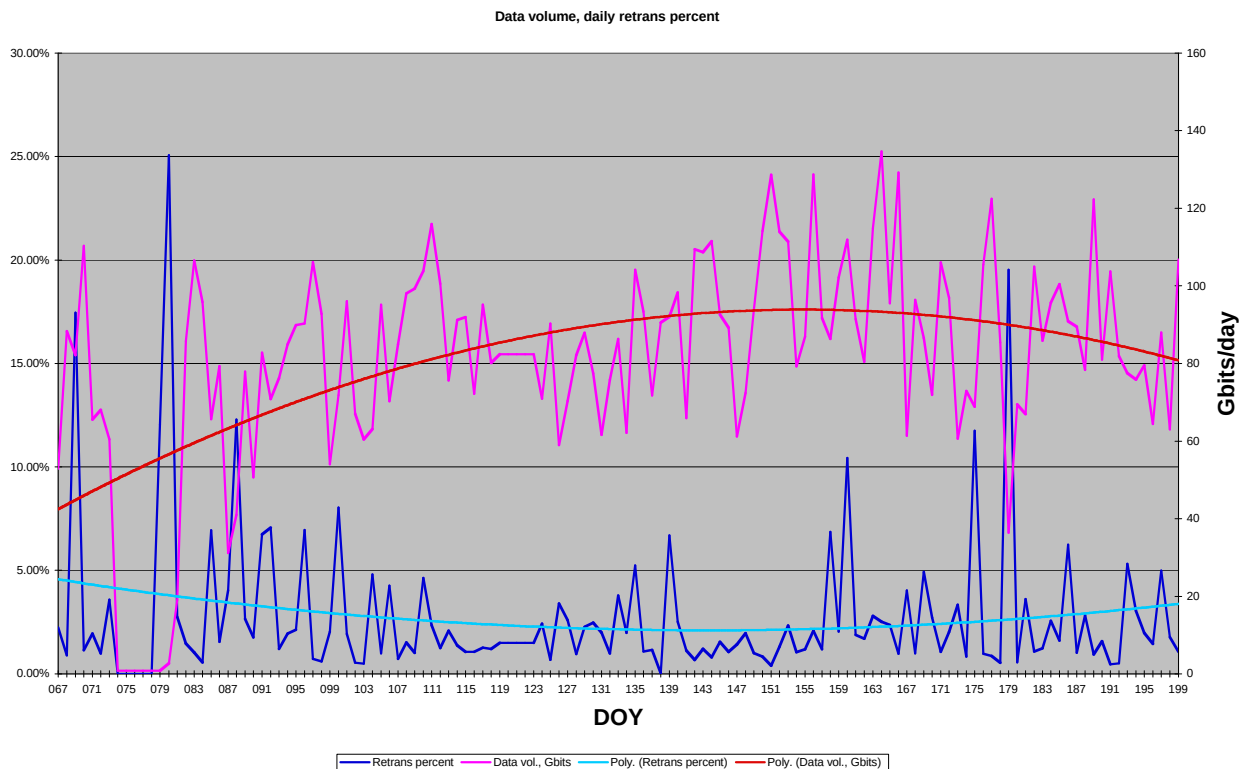
Another important factor in MRO data retransmission involves humans. Given that data can be overwritten onboard in as little as three hours, an ideal system would have retransmission commands being built for nearly every orbit, day and night, weekday and weekend. Currently, the system we use for commanding requires on-console human interaction from an ACE. The human toll of having to man the command-system console around-the-clock in order to radiate retransmission commands is too great to be sustainable for long periods. The current practice is to have an ACE available three times a day on weekdays (beginning, middle, and end of the workday) and twice (morning and afternoon) on weekends. On designated holidays, no retransmission commands are built or radiated.

Another lesser influence on retransmission timing is the nature of the ground process File Delivery Manager (FDM), which re-assembles and distributes the science data products. Current practice is for FDM to keep track of each data product for up to two days, so that if additional data for an incomplete product is received it will be added. After 48 hours, the FDM flushes out whatever results it has on any individual uncompleted product and erases pointers to that product. If retransmission were to cause additional data for a particular product to enter the ground system after this 48-hour period, FDM would interpret the data as a new product. This is obviously to be avoided due to the potential confusion about the status of a particular product downstream of FDM. In practice, this has not turned out to be an operational concern, though rare cases do necessitate some hand processing.



### C. Why we retransmit frames rather than products

There is substantial utility in retransmitting particular missed frames rather than complete data products. A typical partial product before retransmission will often be missing just a few frames. This means that a few seconds of downlink can complete a product sitting as a partial in FDM; to ask for a retransmission of the entire product would take in some cases over an hour. Frame retransmission is much thriftier with time and allows the science teams to make the well-founded assumption that retransmission will occupy less than 3% of the daily data downlink volume, on average. In practice this has borne out to be true, with the average running about 2%. Figure 12 shows values of data returned daily, in Gigabits, accompanied by daily percentage of data that was retransmitted on the same day. Smooth “poly” lines are polynomial fits to the data. The time period is year 2007 from DOY 067 (March 8, 2007) to DOY 195 (July 14, 2007.)



**Figure 12. Retransmission trends**

A second reason for preferring frame-level rather than product-level retransmission is that we can identify frame-level gaps and request retransmission of those missing frames as much as 24 hours earlier than we could identify and request retransmission for an entire missing product. This is due to the DSN 24-hr product latency requirement, i.e., the DSN quite reasonably only commits to delivering data no later than 24 hours after it is initially received by the DSN.

A third reason for doing frame-level retransmission is that the FAR process identifies frame gaps with a latency of under five minutes, thus one can request timely retransmission of missing frames almost as soon as they are identified. If one were to wait until the product was definitively identified as partial by the FDM process there will often be a latency of several hours due to large file sizes, orbit occultations, and scheduled no-coverage periods between DSN stations which lead to products being in the FDM queue for many hours, whether or not they ultimately are completed without retransmission. It's much better to ask for missing pieces as soon as the system identifies them rather than waiting around for the final verdict of “complete” or “partial” which is output by the product-reassembly process in FDM.

Examining just a single instrument, HiRISE, it appears that at higher data rates (4 Megabits/sec) that only about 60% of all their products are initially received as complete.

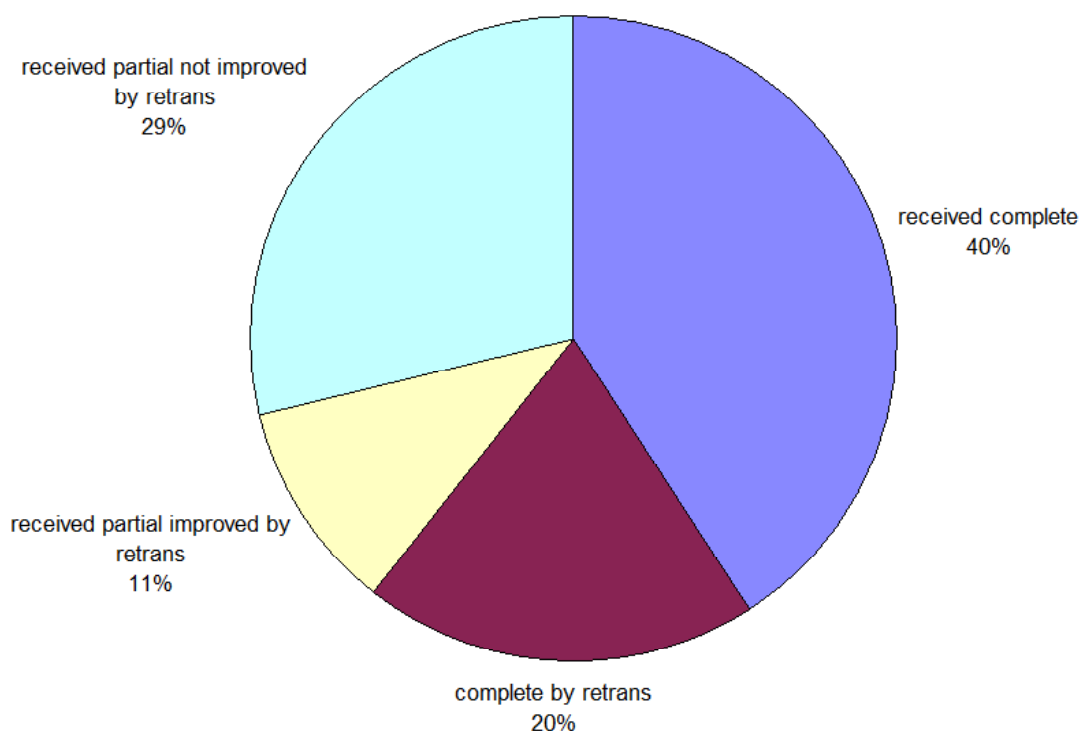
If retransmission was requesting the downlink of entire data products rather than specific frames, that would push the daily percentage of retransmitted data to over 50%, causing a severe degradation in MRO's ability to downlink high data volumes.

The design of MRO's SSR control software makes identification of missing frames, and retransmission of just those frames, quite straightforward. Since many of MRO's science data products are quite sizable (up to 28 Gigabits) and since the downlink performance and the performance of the DSN is quite good, just identifying the few missing pieces of a product is much more elegant than requesting entire products to be retransmitted.

#### D. Effectiveness and usefulness of MRO data retransmission

In an ideal world MRO would never have gaps in its data, and in fact there have been at least two 24-hour periods where MRO had zero gaps and the retransmission software reported "No gaps found – exiting". But space science is rarely performed under ideal conditions. Also, the MRO spacecraft has a known bug which places a gap in data at occasional intervals when using its onboard Turbo encoding scheme; this guarantees that there will be small data gaps in many products when using Turbo encoding. Since Turbo encoding allows the smaller DSN antennas to support higher downlink rates when at large Mars-to-Earth ranges, there's an obvious tradeoff: accept occasional small gaps in data products, filling them in with frame-level retransmission where possible, or use lower data downlink rates so as to have gap-free products. The Turbo anomaly is not data rate dependent, so we would still miss frames at **any** downlink rate using Turbo. And "lower" data rates would imply not just a little lower, but perhaps **half** the rate of using Turbo.

For science team planning and workflow purposes, experience has shown that filling a gap is far more effective than getting a new product. Science processing pipelines can and do become overwhelmed when dealing with large numbers of partial products, especially since the teams will probably get multiple versions of these products as additional data comes in. Getting an unambiguously complete product really makes life easier for the science data processing teams. See Fig. 13 for a representation of how retransmission has benefited the MRO project at the product level.



**Figure 13. Impact of retransmission on product completeness (HiRISE Products)**

## E. Limitations of the MRO data retransmission system

As mentioned above, when the onboard overwrite time of data products is short (in the range of four hours or less), retransmission becomes an exercise in frustration. There are feedback tendencies in the system: if a large gap occurs because of Earth weather or other ground issues and a large retransmission request is uplinked while the onboard SSR is quite full, the result will be to make the framed space in the SSR be even fuller, shortening the effective time available to retransmit missed frames, possibly resulting in the undesirable outcome of using a lot of time to ask for something that isn't there any more, or if it is still there, causing a further backlog of data onboard, possibly even causing data to overflow the SSR. Everything works better when there's a bit of headroom for retransmission. See Fig. 11 above.

The Frame Accountability Reports (FARs) are an excellent means of assessing what data has been missed in the downlink process, but they do occasionally give inaccurate results. These inaccuracies usually are the result of DSN hardware or software anomalies that may require restarting multiple processes. When DSN processes are restarted mid-track, the FARs can become corrupted in certain circumstances and either flag gaps that aren't there, or perhaps even worse, miss gaps that **are** there. In practice the FARs have been quite reliable, but definite evolution has occurred due to various unanticipated circumstances that were not envisioned in testing. Both the FAR generation software and the "rcgen" software that ingests the FARs have had to have software updates to patch problems seen in flight, and there are probably still other things yet to learn and patch.

When very large gaps, in excess of one million frames, occur, our retransmission system is unsuited to trying to fill in those very large gaps. For instance, there have been a couple of occasions where a multi-hour DSN track has been lost due to equipment problems on an antenna or at an entire DSN complex. It is unrealistic to use retransmission to fill in such very large gaps. The practical limit appears to be approximately one orbit's worth of data; successful retransmissions of one-orbit gaps (around 70 minutes worth of downlink) have been done numerous times.

Due to the limited space of MRO's SSR, an ideal situation would be to request retransmission of data *immediately* whenever gaps were identified. But as discussed above, this would require ACEs to be on-console 24x7, which is an inefficient use of human resources. Limited auto-commanding for retransmission seems like a good technical solution to this problem; some developments currently underway show promise in being able to do unattended retransmission commanding while still complying with project and other security requirements.

MRO has found that at higher data rates (3.3 Megabits/sec and above) that there are occasional "ratty" periods where the data received has many small gaps in it. These "ratty" periods often span only a time period of ten minutes or so, but due to the high data rates have many thousands of gaps. It's a difficult balance to adjust the retransmission software to aggregate these gaps and yet not overwhelm the system with large amounts of already-received data. The alternative (asking for each individual gap) leads to commands which have many thousands of instruction lines, and this runs into limitations of the sequencing software to process more than a few thousand instruction lines, causing the retransmission and sequence software to reject what it sees as excess instructions. Of course, the rejected portion will be the newest gaps identified, since for several good reasons the command-building software goes from oldest to newest of gaps; unfortunately these are the gaps that are most-likely still to be resident onboard!

Experience has shown that there are just a LOT of places where things can go wrong in the retransmission chain: ground file systems full or not available, corrupted FARs, mail notification failures to ACEs, gap(s) exceeding configured limits, ground workstation process limits exceeded, incorrect downlink encoding called for, DSN problems during the retransmission, typos in the crontab file, and more. The saving grace is that retransmission is not mission-critical but rather is an enhancement to improve data quality

## F. Optimization strategies

Early on in the MRO mission it was realized that reserving a certain proportion of SSR frame space where data products are assembled would allow greater times until data overwrite. This reserve has been set at 20% of total frame space for most of the mission to date. With higher data rates, it was initially anticipated that retransmission would be discontinued; however, it has proven so valuable that new strategies are being devised to allow effective retransmission even at the highest data rates. These strategies inevitably require reconfiguration of the SSR, which requires science teams to drain out their observations so as not to lose them. This takes time and a fair amount of planning so is done as seldom as possible.

The tools used for retransmission command operations have been in a fairly constant state of evaluation and improvement since before PSP. Many of the modifications have been made to accommodate things that were not

known and were unanticipated in testing. This is the nature of any form of exploring; the trick is in being flexible enough to accommodate needed changes while at the same time retaining a higher-level sense of direction and control. The authors feel that this balance has been achieved admirably on MRO for the retransmission software and procedures.

#### **G. Summary of lessons learned concerning MRO data retransmission:**

- ◆ Frame-level retransmission works and is very worthwhile!
- ◆ Visibility into the retransmission process is vital (the processes can't be too "black-box")
- ◆ Filling a gap in a science data product is **much** more important to the science teams than getting more new (gappy) products
- ◆ Having a lightweight, flexible, and easily-modifiable system is a big bonus when doing things previous projects have not done, giving you agility to quickly adapt and optimize software, processes, and procedures. The core software and procedures, however, must be well thought-out in the beginning and have at their nucleus well-crafted requirements.
- ◆ Automation, especially scripted e-mail notifications of status, is critically important to keep situational awareness throughout the team and to reduce human workloads.

## **VII. Conclusion**

To address challenges inherent in handling unprecedented telemetry rates and volumes for a planetary mission, the MRO project made a strategic investment in the area of end-to-end science data accountability. A new team, the end-to-end Data Accountability Team (eeDAT), was established specifically to address these challenges. The team was formed early enough in the project development life-cycle to allow for development of a novel telemetry product-level accountability system, and early enough to influence the design of the flight and ground software to provide status telemetry from key points in the end-to-end data flow.

The primary science data collection phase for the mission began in November of 2006. Since that time, an average of over 10,000 science products per month have been generated. For each and every product, predictions were generated, progress was monitored using new web-based tools, and periodic performance statistics been produced. A semi-automated, frame-level retransmission process was able to improve the completeness of 31% of products that were originally missing or incomplete due to losses in downlink, fully completing 20% of these products. The eeDAT team investigated the cause of all products that were still missing after retransmission, and documented these causes in a web-accessible database. The team was also able to identify and categorize the causes of various types of data losses, including both large contiguous losses, and periods of scattered small losses referred to as ratty periods. By working with other operations teams, as well as flight and ground system development teams, many of the root causes of data loss have been corrected. Interestingly, some of these causes had been present in our multimission systems for some time, but had gone previously unnoticed. The team continues to collect and analyze performance data, and expects to continue to make improvements in the data systems, resulting in improved science product quality.

The MRO investment has clearly paid off by providing significant improvements in science data volume and quality. As future missions trend toward higher and higher data rates and volumes, and as product-level telemetry becomes more commonplace, they will be able to build upon the MRO end-to-end accountability architecture and benefit from many lessons learned.

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